

BRAC University
Department of Computer Science and Engineering
CSE420:Compiler Design

Examination: Quiz 3
Time: 20 min

Semester: Fall 25
Marks: 15

Name:		ID:	Section:
-------	--	-----	----------

1. Using the given syntax-directed translation (SDT), draw the parse tree, evaluate all relevant node attributes, and generate the three-address code (3AC) for the input string:
L [c] [z] = a ;

Let L denote a 5 x 6 array of integers, and let c, z, and a all denote integers. Then, the type of L is array(5, array(6, integer)). Its width w is 120, assuming that the width of an integer is 4. The type of L[c] is array(6, integer), of width = 24. The type of L[c] [z] is integer.

```

S → id = E ;    { gen( top.get(id.lexeme) '=' E.addr); }

      | L = E ;    { gen( L.addr.base '[' L.addr ')' '=' E.addr); }

E → E1 + E2    { E.addr = new Temp();
                  gen( E.addr '=' E1.addr '+' E2.addr); }

      | id          { E.addr = top.get(id.lexeme); }

      | L            { E.addr = new Temp();
                  gen( E.addr '=' L.array.base '[' L.addr ')' ); }

L → id [ E ]    { L.array = top.get(id.lexeme);
                  L.type = L.array.type.elem;
                  L.addr = new Temp();
                  gen( L.addr '=' E.addr '*' L.type.width); }

      | L1 [ E ]    { L.array = L1.array;
                  L.type = L1.type.elem;
                  t = new Temp();
                  L.addr = new Temp();
                  gen( t '=' E.addr '*' L.type.width); }
                  gen( L.addr '=' L1.addr '+' t ); }

```